

Metaheuristic Hyperparameter Optimization for Intraday Financial Trend Classification: A Benchmark Across Classifier Backbones

Abstract

Short-horizon trend classification in futures markets is vulnerable to non-stationarity, weak signal-to-noise ratios, and temporal information leakage. An earlier conference precursor combined technical indicators, Information Gain feature selection, and genetic-algorithm tuning, but used randomized cross-validation, a single metaheuristic, and no economic assessment. This study advances that research line through an auditable benchmark on 15,057 five-minute B3 Mini-Index futures bars. Four model backbones (multilayer perceptron, random forest, support vector machine, and 1D convolutional neural network) are optimized by random search, genetic algorithm, particle swarm optimization, differential evolution, and grey wolf optimization under an identical 1,000-evaluation budget. A chronological 60/20/20 train/validation/test partition prevents look-ahead leakage. The study contrasts two validation objectives: a compound $0.60 \text{ MCC} + 0.40 F_1$ fitness and a weighted training/validation accuracy fitness. The common, locked test block is used only after selection for predictive and trading evaluation. Results are reported transparently on the three seeds available in every model-optimizer cell. The direct fitness comparison is descriptive; complementary Friedman and Holm-adjusted Wilcoxon analyses compare models within each protocol across matched optimizer-seed blocks. The study provides a reproducible design for separating the selection criterion from the reported metric and for jointly examining predictive and economic consequences of metaheuristic tuning.

Keywords: Financial time series · Metaheuristic optimization · Hyperparameter tuning · Matthews correlation coefficient · Temporal validation · Intraday futures

1 Introduction

Financial time-series forecasting remains a challenging computational problem because market data exhibit non-stationarity, nonlinear dependence, structural changes, and substantial noise [1, 2]. These characteristics become particularly relevant in short-horizon trading, where predictive relationships may weaken or change as market conditions evolve. Among traded financial instruments, futures contracts provide an important setting for this problem because they are widely used for both risk hedging

and speculative exposure [3, 4]. In Brazil, the Mini-Index futures contract (symbol: WIN), traded on Brasil Bolsa Balcão (B3) and referenced to the Ibovespa index, is highly liquid and extensively used in intraday operations, making it a relevant environment for computational financial research [5, 6]. Predicting directional movements in such markets is therefore scientifically demanding and requires methods capable of extracting useful information from noisy and temporally evolving observations.

Traditional financial forecasting has relied on statistical models such as autoregressive integrated moving average (ARIMA) techniques [7]. As finance and computer science increasingly converged under the computational intelligence paradigm [8], machine-learning models became widely investigated for financial prediction and classification. Supervised approaches—including Random Forests (RF) [9], Support Vector Machines (SVM) [10], Multilayer Perceptrons (MLP) [11], and 1D Convolutional Neural Networks (1D-CNN) [12]—can capture nonlinear relationships between market variables and future price movements. These models are commonly supplied with technical indicators derived from price, volume, momentum, and volatility. However, large indicator sets may introduce redundant or weakly informative variables, increasing model complexity and potentially degrading generalization [13, 14]. Feature-selection techniques therefore provide an important mechanism for identifying compact and informative predictor subsets [4, 15].

Beyond feature selection, predictive performance also depends strongly on hyperparameter configuration [16]. Neural and machine-learning models contain structural, regularization, and training parameters that jointly define mixed discrete–continuous and often non-convex search spaces. Gradient-based learning algorithms optimize model parameters once an architecture is defined, but they do not directly solve this outer hyperparameter-search problem when discrete and non-differentiable choices are involved [17]. Metaheuristic Optimization Algorithms (MOAs) provide an alternative by exploring candidate configurations through stochastic search. Evolutionary approaches such as Genetic Algorithms (GA) [18] and Differential Evolution (DE) [19], together with swarm-based methods such as Particle Swarm Optimization (PSO) [20] and Grey Wolf Optimizer (GWO) [21], have consequently been applied to automated model tuning across neural computing problems [22, 23].

Despite the increasing use of metaheuristic optimization in predictive modelling, comparisons between algorithms remain strongly influenced by experimental design. Three methodological limitations are particularly relevant in financial time-series applications:

1. **Unequal computational budgets:** Metaheuristics are commonly expressed using different units such as generations, iterations, populations, particles, or wolves. As a result, apparently similar experimental configurations may correspond to substantially different numbers of objective-function evaluations. Without a common evaluation budget, observed differences may reflect unequal search opportunities rather than differences in optimizer behaviour.
2. **Temporal data leakage:** Randomized data partitioning and conventional shuffled cross-validation can violate the chronological structure of financial observations

[2]. When future observations contribute directly or indirectly to model selection, validation results may provide an overly optimistic estimate of out-of-sample generalization.

3. **Dependence on the optimization objective:** Hyperparameter-search outcomes depend not only on the optimizer but also on the metric used to rank candidate configurations. Accuracy alone may provide an incomplete assessment of classification quality, particularly when prediction errors or class distributions are unbalanced [24]. Moreover, improvements in predictive metrics do not necessarily translate into economically useful trading behaviour once transaction costs and signal frequency are considered.

To address these limitations, this paper presents a controlled multi-optimizer benchmark for intraday trend classification using 5-minute B3 Mini-Index futures data. A compact seven-indicator subset previously identified through Information Gain is used as a fixed input representation, while five optimization strategies—Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO)—are evaluated across four classifier families: MLP, SVM, RF, and 1D-CNN. All optimizers operate under the same objective-function evaluation budget of $N_{\text{eval}} = 1,000$ evaluations per seed, and temporal ordering is preserved through chronological validation with an isolated final test block. The study further contrasts a compound MCC-driven objective ($0.60 \text{ MCC} + 0.40 F_1$) with accuracy-based optimization protocols. Predictive performance is complemented by convergence analysis, non-parametric statistical testing, and out-of-sample economic evaluation under transaction costs.

The main contribution of this study is the development of a controlled experimental framework that disentangles the effects of optimizer family, predictive model, validation objective, and computational budget in intraday financial classification. Rather than evaluating metaheuristics under isolated or algorithm-specific settings, the proposed benchmark places five optimization strategies and four predictive model families under the same temporal data partitions, the same feature representation, and an equal number of objective-function evaluations. By combining predictive metrics, convergence behaviour, matched statistical testing, and economic evaluation under transaction costs, the study provides a more complete assessment of optimizer robustness and practical relevance. In this sense, the work advances previous Brazilian futures research beyond feature selection and single-optimizer tuning toward a reproducible comparative methodology for studying model-optimizer interactions under realistic temporal and computational constraints.

The remainder of this manuscript is organized as follows. Section 2 presents the related literature and establishes the research gap. Section 3 describes the dataset, feature selection, predictive models, optimization algorithms, search spaces, and experimental protocol. Section 4 reports the predictive results, convergence behaviour, statistical analyses, and economic evaluation. Section 5 discusses the findings, methodological implications, and limitations. Finally, Section 6 summarizes the main conclusions and outlines future research directions.

2 Systematic Review of Literature

The intersection of computational intelligence, metaheuristic optimization, and financial time-series forecasting has generated an extensive body of literature over the past decade [8, 17, 25]. Comprehensive surveys have categorized machine learning applications in financial prediction [8, 25], systematically reviewed metaheuristic search paradigms such as Particle Swarm Optimization [26, 27] and Grey Wolf Optimizer [21], and benchmarked neural forecasting architectures under computational constraints [16, 28]. However, despite substantial published progress in premier neural computing journals [12, 22, 23, 29], existing reviews and empirical benchmarks have not systematically evaluated the interaction between equal-budget metaheuristic tuning, temporal validation integrity, and out-of-sample trading performance. This section presents a Systematic Review of Literature (SRL) guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) protocol, examining prior work across feature selection, model backbones, metaheuristic search, and validation protocols, synthesizing current state-of-the-art contributions and formalizing the three critical methodological gaps that motivate our benchmark framework.

2.1 Literature Selection Protocol and Research Questions

To ensure methodological rigor and eliminate reporting bias, relevant literature was retrieved following an explicit PRISMA search protocol across major scholarly databases: IEEE Xplore, Scopus, Web of Science, and SpringerLink (specifically targeting *Neural Computing and Applications*). The search was executed using the following structured Boolean query:

- ("metaheuristic" OR "genetic algorithm" OR "particle swarm" OR "grey wolf")
- AND ("feature selection" OR "technical indicators" OR "Information Gain")
- AND ("futures market" OR "intraday forecasting" OR "trend classification")

The review was bounded to peer-reviewed journal articles and top-tier conference proceedings published between 2019 and 2026. Candidate studies were screened against explicit eligibility criteria:

- **Inclusion Criteria:** (i) evaluation of automated feature selection or metaheuristic hyperparameter tuning on financial or complex time-series classification/forecasting; (ii) explicit disclosure of validation protocols and search space boundaries; and (iii) reporting of empirical predictive or out-of-sample financial metrics.
- **Exclusion Criteria:** Studies lacking reproducible experimental parameters, omitting candidate evaluation budget details, or failing to disclose optimization constraints were excluded.

Figure 1 depicts the complete PRISMA literature selection workflow. The systematic review is organized around three central Research Questions (RQs):

1. **RQ1 (Budget Parity):** How do stochastic metaheuristic search operators (RS, GA, PSO, DE, GWO) compare across distinct classifier backbones (RF, SVM, MLP, 1D-CNN) when restricted to a strictly equal candidate evaluation budget cap ($N_{\text{eval}} = 1,000$)?
2. **RQ2 (Temporal Leakage):** What is the quantitative impact on out-of-sample generalization when comparing chronological sequential temporal splits against randomized cross-validation on high-frequency financial time series?
3. **RQ3 (Economic Alignment):** How effectively do classification-based validation fitness objectives (e.g., compound MCC/ F_1 vs. weighted accuracy) align with out-of-sample intraday trading metrics (Net Return %, Sharpe ratio, Maximum Drawdown) under realistic exchange transaction costs?

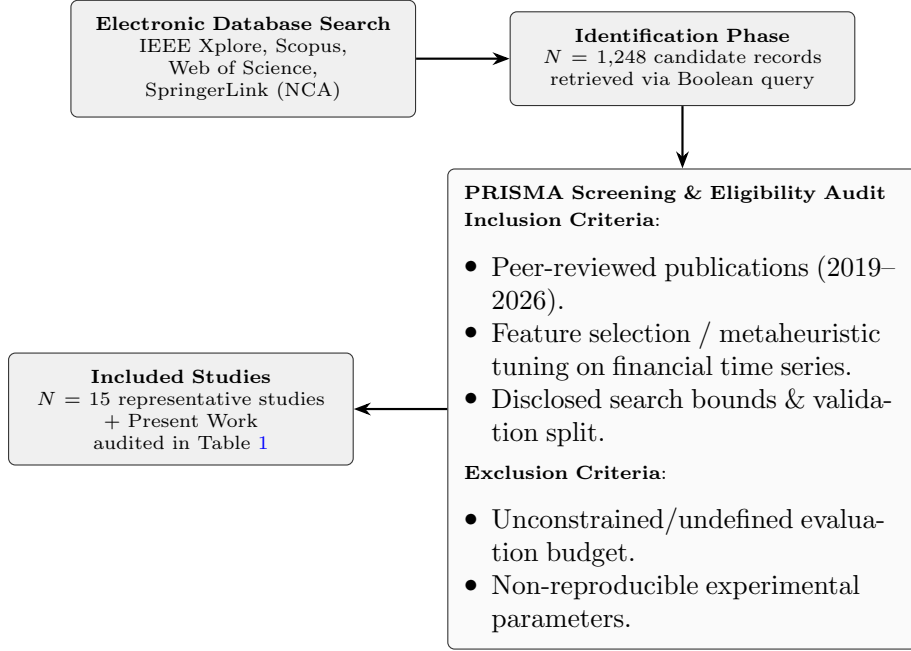


Fig. 1 PRISMA literature selection protocol flowchart mapping database retrieval, eligibility criteria filtering, and final synthesis of 15 representative studies audited in Table 1.

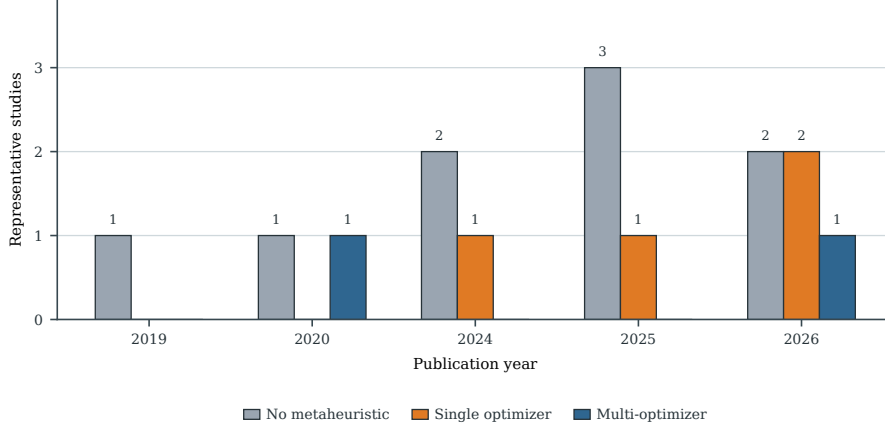


Fig. 2 Distribution of the 15 representative prior studies included in the literature synthesis, grouped by publication year and the scope of optimizer comparison. A study is assigned to one category according to its primary experimental design; the present work is excluded. The limited number of multi-optimizer studies motivates the controlled benchmark developed in this paper.

2.2 Classifier Backbones and Input Representations

The quality and information density of the input representation are central determinants of supervised intraday trend prediction [13]. Open, high, low, close, and volume observations encode the market state, but short-horizon bars are also affected by microstructural noise, regime shifts, and high variance. For this reason, financial machine-learning studies commonly transform raw bar data into technical descriptors representing complementary aspects of price behaviour: momentum, trend, dispersion, and volatility [29–31]. Such representations do not create information; rather, they provide a structured hypothesis space through which a learner can test whether recent price and volume dynamics carry directional content.

The expansion of the indicator set must nevertheless be controlled. Indicators derived from common prices and overlapping lookback windows are often strongly correlated, so a large technical library can amplify redundancy, increase estimation variance, and make model ranking sensitive to incidental patterns in the development sample [14, 15]. Feature selection is therefore a component of model design rather than a preliminary convenience: it reduces the dimensionality exposed to subsequent learning and hyperparameter search, while retaining descriptors with a demonstrable association with the target. Information Gain (IG) is a transparent filter criterion for this purpose, ranking candidate indicators by their reduction of class uncertainty [14, 32]. Although a univariate ranking does not guarantee an optimal multivariate subset, its low computational cost and auditability make it appropriate for constructing a compact technical representation. The mathematical formulation and empirical application of IG in the present benchmark are detailed in Section 3.3.

The literature spans several classifier families whose distinct inductive biases make a single preferred backbone unlikely across all market representations:

- **Ensemble decision trees**, represented by Random Forest (RF), aggregate decorrelated trees built from bootstrap samples and randomly selected feature subsets. Their partition-based structure accommodates non-linear interactions and can be relatively robust to noisy descriptors [9].
- **Kernel margin classifiers**, represented by Support Vector Machines (SVM), seek separating hyperplanes with a maximum-margin principle. Kernel mappings, including the radial-basis-function kernel, permit non-linear decision surfaces while retaining a well-defined optimization objective [10].
- **Feedforward neural networks**, represented by Multilayer Perceptrons (MLP), learn non-linear mappings through stacked fully connected layers and backpropagation. Their expressive capacity is accompanied by sensitivity to architectural and regularization choices [11].
- **One-dimensional convolutional neural networks** (1D-CNN) apply shared local filters to ordered input windows. This construction can capture short-range patterns while using parameter sharing to constrain the number of learned weights [12, 28].

These families therefore represent different hypotheses about how directional information is encoded. Their comparative use is more informative than selecting a single algorithm a priori, particularly because changes in feature representation and hyperparameterization can alter the observed ranking of classifiers [9, 33]. The specific representations, search domains, and fitted architectures adopted in this study are defined in Section 3.

2.3 Metaheuristic Optimization Paradigms

Hyperparameter selection is naturally formulated as a black-box optimization problem. Let $\theta \in \Omega$ denote a feasible model configuration, where Ω can contain continuous, integer, categorical, and conditional variables. A validation criterion $f(\theta)$ is obtained only after fitting and evaluating the corresponding learning pipeline, and the search objective is

$$\theta^* = \arg \max_{\theta \in \Omega} f(\theta). \quad (1)$$

Unlike weight training, this objective need not be differentiable with respect to θ . Tree depth, number of hidden units, kernel choices, and regularization settings may also be discrete or conditionally active. Population-based metaheuristics are consequently attractive because they navigate such mixed spaces without gradient information and can balance broad exploration against local refinement [16, 17, 21].

In this context, a “candidate evaluation” should mean one complete configuration assessment under an unchanged validation protocol. This distinction is important: iterations and generations are internal counters, whereas the number of evaluated candidates is the common computational unit through which heterogeneous search procedures can be compared.

Figure 3 provides a compact taxonomy of the learning and search families considered in this literature. It distinguishes the classifier hypothesis spaces from the mechanisms used to select their configurations, while retaining the shared objective of validation-based model selection.

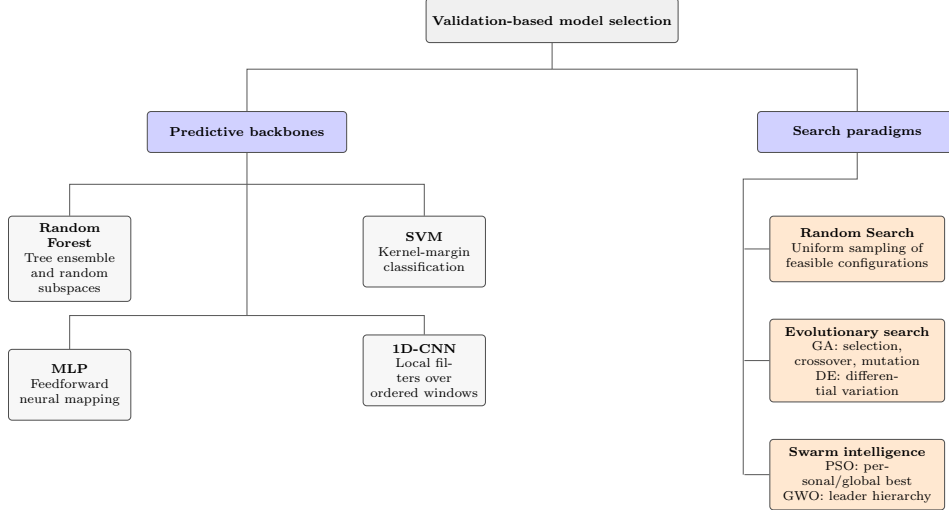


Fig. 3 Conceptual taxonomy linking predictive backbones with hyperparameter-search strategies in the reviewed literature. The left branch denotes alternative classifier hypothesis spaces, whereas the right branch distinguishes unguided sampling, evolutionary search, and swarm intelligence. The precise search domains and experimental controls are specified in Section 3.

Evolutionary algorithms maintain a population of candidate configurations and improve it through selection and variation. Genetic Algorithms (GA) emulate selection, crossover, and mutation: configurations with stronger fitness are preferentially retained or recombined, while mutation introduces new material into the population [16, 18]. This representation-agnostic cycle is well suited to mixed hyperparameter spaces after appropriate encoding and constraint handling.

Differential Evolution (DE) is also population based, but creates candidate displacements from scaled differences between population members. A canonical donor-vector form is

$$\mathbf{v}_i^t = \mathbf{x}_{r_1}^t + F (\mathbf{x}_{r_2}^t - \mathbf{x}_{r_3}^t), \quad (2)$$

where r_1 , r_2 , and r_3 index distinct population members and F scales the differential displacement. Crossover then combines the donor with a target vector, and selection retains the better candidate [19]. The use of population differences gives DE a self-adaptive search scale, although its practical behaviour still depends on encoding, bounds, and the chosen mutation and crossover strategy.

Swarm methods update candidates by combining individual experience with population-level information. In Particle Swarm Optimization (PSO), each particle has a position $\mathbf{x}_i^t$, a velocity $\mathbf{v}_i^t$, a personal best position $\mathbf{p}_i^t$, and an incumbent swarm-best position $\mathbf{g}^t$. Its standard update is

$$\mathbf{v}_i^{t+1} = \omega \mathbf{v}_i^t + c_1 r_1 (\mathbf{p}_i^t - \mathbf{x}_i^t) + c_2 r_2 (\mathbf{g}^t - \mathbf{x}_i^t), \quad (3)$$

$$\mathbf{x}_i^{t+1} = \mathbf{x}_i^t + \mathbf{v}_i^{t+1}, \quad (4)$$

where ω controls inertia, c_1 and c_2 weight cognitive and social attraction, and r_1, r_2 are random variates. This formulation makes the exploration–exploitation trade-off explicit and is the foundation for the broad family of PSO variants reviewed by Gad [20, 26].

Grey Wolf Optimizer (GWO) uses a different collective metaphor. Candidate solutions are ranked into leading α , β , and δ wolves, while the remaining wolves update their positions by encircling and approaching these leaders. Its control schedule progressively shifts the search from exploration to exploitation [21, 22]. Despite distinct metaphors, PSO and GWO share the same benchmarking requirement: each update must be translated into feasible configurations and assessed under the same validation rule.

Random Search (RS) is an indispensable baseline because it samples feasible configurations without an adaptive search trajectory. It establishes whether the additional mechanisms of evolutionary or swarm search improve solution quality beyond unguided sampling from the same search space [34]. A credible comparison must therefore control the parameter domain, candidate-evaluation count, validation data, random replication, and reporting metrics. Equalizing only the nominal number of generations or iterations is insufficient when algorithms evaluate different numbers of candidates per update.

The comparative evidence also highlights recurring weaknesses: unequal computational budgets can confound search quality with search effort; randomized validation can compromise temporal integrity; and classification-only reporting may not establish economic value after transaction costs [11, 12, 29]. These issues are synthesized in Section 2.4; the concrete controls used by the present benchmark are specified only in Section 3.

2.4 Comparative Synthesis of Representative Studies

To establish the state of the art and pinpoint persistent methodological gaps, Table 1 synthesizes 15 representative published studies from 2019 to 2026 alongside the present benchmark work. Studies are evaluated across six technical dimensions: Study & Year, Application Domain & Data, Model Backbones, Optimization & Feature Paradigms, Validation Protocol, and Identified Methodological Gap.

As synthesized in Table 1, three primary methodological flaws permeate the existing literature on metaheuristic financial forecasting:

1. *The Budget Inequality Flaw*: Prior studies routinely compare metaheuristics without enforcing a strict candidate evaluation cap (N_{eval}). Because stochastic optimization performance scales with function evaluations, superiority claims in unconstrained studies often reflect unequal computational budgets rather than algorithmic search efficiency [11, 28].
2. *The Temporal Data Leakage Flaw*: Financial machine learning research frequently relies on randomized K -fold cross-validation or data shuffling [2, 35]. Shuffling sequential intraday price bars allows future market information to contaminate historical training folds, yielding artificially inflated predictive accuracy scores that collapse during live execution.

3. *The Classification-Economic Disconnect Flaw*: Hyperparameter tuning is overwhelmingly conducted against raw accuracy or cross-entropy loss [24, 32]. In high-frequency futures trading, raw accuracy ignores class imbalance and trade execution costs. A classifier achieving high nominal accuracy can incur net financial losses once exchange transaction fees and slippage are deducted.

Table 1 Systematic Literature Review comparison matrix auditing representative studies in financial machine learning, metaheuristic hyperparameter tuning, and temporal validation protocols against identified methodological gaps.

Study & Year	Domain & Data	Models	Optimization & Features	Validation Protocol	Methodological Gap
Nabipour et al. (2020) [9]	Stock index trend	RF, SVM, ANN, XGBoost	Technical indicators	Train/test split	Absence of metaheuristic hyperparameter tuning.
Jovanovic et al. (2024) [16]	Crop-yield prediction	Weight-agnostic neural networks	Metaheuristic tuning	Train/test evaluation	Non-financial domain; no equal-budget financial benchmark.
Naik & Mohan (2019) [10]	Equities (NSE India)	ANN classifier	Boruta feature selection	Train/test split	Static parameters; no metaheuristics.
Ecer et al. (2020) [11]	Financial trend	MLP classifier	GA and PSO tuning	K -fold CV	Unconstrained budget; single classifier.
Billah et al. (2024) [29]	Intraday equities	Heuristic models	Moving average rules	Standard split	Absence of neural hyperparameter tuning.
Bareket & Párv (2024) [36]	Surge detection	Adaptive ML	Technical indicators	Medium-term split	Evaluation restricted to static features.
Bandgar et al. (2025) [12]	Financial forecasting	Ensembles	Default tuning	Random split	No evaluation budget cap; data leakage.
Rahim et al. (2025) [37]	5-min futures	Single neural model	Single GA optimization	Train/test split	Single GA; no multi-swarm benchmark.
Dhingra et al. (2025) [28]	Solar PV forecasting	LSTM, CNN-LSTM	RAdam, Lookahead, Adam	Sequential split	Non-financial domain; stochastic optimizers omitted.
Mutinda & Yong (2026) [33]	Financial market	Ensembles	SEDT feature selection	K -fold CV	Feature selection without tuning.
Simatupang et al. (2026) [35]	Stock index (IDX)	Bayesian RF	Bayesian optimization	Random K -fold CV	Random CV causes temporal data leakage.
Al Bannoud et al. (2026) [22]	Kinetics time series	ANN-ODE surrogate	GWO, PSO, GA, ABC	Multi-partition	Non-financial; unconstrained search budget.
Bashir et al. (2026) [23]	Equities (NSE India)	Bagging	(1+1) Evolutionary Algo	Train/test split	Single EA evaluated without budget cap.
[Anonymized Precursor A]	Intraday B3 futures	Ensemble Random Forest	InfoGain + Default RF	Sequential 60/20/20	Untuned baseline classifier without metaheuristics.
[Anonymized Precursor B]	Intraday B3 futures	MLP, SVM, RF	InfoGain + Single GA	5-fold CV	Single GA; unconstrained 1,000-generation budget.
Present Work	Intraday B3 futures	MLP, SVM, RF, 1D-CNN	InfoGain + RS, GA, PSO, DE, GWO	Sequential 60/20/20	Equal budget ($N_{eval} = 1,000$), 3 seeds, fee backtest.

2.5 Research Positioning and Study Contribution

The literature synthesis reveals that prior studies have typically addressed only subsets of the experimental problem considered here. Existing financial forecasting research has investigated feature selection, individual predictive models, or isolated metaheuristic tuning strategies, but these components are rarely evaluated simultaneously under

common temporal and computational controls. The precursor studies in this research line first established the value of Information Gain feature selection for B3 Mini-Index futures and subsequently explored Genetic Algorithm-based hyperparameter tuning. The present work moves beyond these isolated stages by treating model selection and optimizer behaviour as components of a unified comparative benchmark.

The methodological extension is therefore not limited to increasing the number of models or optimization algorithms. The proposed framework changes the experimental control structure by combining four heterogeneous predictive backbones with five search strategies under a common feature representation, identical temporal partitions, and an equal objective-function evaluation budget. It further evaluates how optimizer behaviour changes under alternative validation objectives and complements predictive metrics with convergence analysis, matched statistical testing, and economic evaluation under transaction costs.

This design allows the study to distinguish performance differences associated with optimizer behaviour from those induced by model architecture, validation objective, or unequal search effort. Accordingly, the contribution is positioned as a controlled and reproducible model-optimizer benchmark for intraday financial classification rather than as an incremental extension of a single metaheuristic method.

3 Materials and Methods

This section presents the methodology adopted in this study. Figure 4 provides an overview of the methodological flow adopted in this study. The diagram highlights each step of the process, illustrating the flow from data acquisition to model optimization and evaluation.

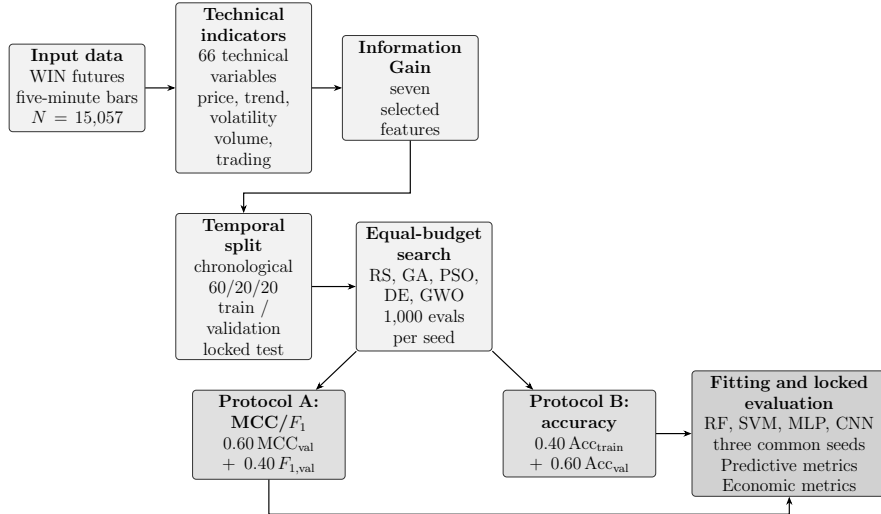


Fig. 4 Overview of the proposed methodology. The benchmark reuses the precursor’s fixed Information-Gain representation while enforcing a chronological split, equal optimizer budget, and two separated validation objectives.

3.1 Input Data

The data used in this study were obtained from BovDbV2 [5], an extension of the original BovDb [6]. The database provides publicly available and pre-processed B3 market data for research purposes, covering listed assets from 1995 to 2024. BovDbV2 additionally includes intraday observations aggregated into 5-minute intervals for the first half of 2024, enabling the analysis of short-horizon market behaviour.

For the present study, the analysis focuses on Mini Ibovespa Futures (WIN) contracts traded between January and June 2024. The selected contracts correspond to the most actively traded maturities over the analyzed period: WING24 and WINJ24 in January–February, WINJ24 and WINM24 in March–April, and WINM24 and WINQ24 in May–June. After preprocessing and construction of the modelling dataset, the final sample contains 15,057 chronological observations.

The explanatory variables are technical indicators computed from the intraday price and volume series. The prediction target is a binary market-trend label, where 0 denotes a downtrend and 1 denotes an uptrend. These labels were generated using the peak-and-trough identification procedure established in previous work and retained here to preserve consistency with the earlier experimental pipeline.

All observations are kept in their original chronological order. The first 9,034 observations are assigned to the training set, the following 3,011 observations to the validation set, and the final 3,012 observations to the locked test set. No random shuffling is performed in the primary holdout protocol, ensuring that the test block remains temporally subsequent to the data used for model fitting and hyperparameter selection.

3.2 Generation of Technical Analysis Indicators

Technical indicators were built from the intraday dataset to capture short-term price changes. Key metrics—Simple Moving Average (SMA), Exponential Moving Average (EMA), and Standard Deviation—were calculated using short time windows (3, 5, 7, and 9 periods) to enhance sensitivity to rapid market shifts, where significant moves can happen in minutes [30].

To capture changes in trend slope and momentum, attributes derived from the differences between calculated moving averages were also constructed. For example, the difference between the 7-period SMA and the 3-period SMA is defined in Eq. (1) as:

$$\text{SMA}_{7-3} = \text{SMA}_7 - \text{SMA}_3, \quad (5)$$

A central preprocessing step consisted of normalizing the technical indicators [13], as well as the opening and closing prices, in order to place all variables on a common numerical scale. For each 5-minute intraday window, a min–max–type scaling was applied, in which each feature was transformed according to Eq. (2):

$$\text{Feature}_{\text{norm}} = \frac{\text{Feature} - \text{Low}}{\text{High} - \text{Low}}. \quad (6)$$

Where *High* and *Low* correspond to the maximum and minimum prices observed within the same 5-minute interval, and *Feature* denotes any technical indicator or

price-based variable being normalized. This transformation maps the features predominantly to the interval $[0, 1]$, while preserving their relative positions within the intraday price range, thereby reducing the impact of heterogeneous attribute scales and facilitating the learning of machine learning models. This normalization procedure, combined with historical information on open, high, low, average, close, bid, ask, trade volume, and number of shares, results in a total of 66 constructed features.

3.3 Feature Selection Method

The predictive pipeline incorporates a structured feature selection stage based on technical analysis indicators. Following the approach proposed in [5], multiple selection strategies are evaluated to identify the most informative feature subsets, thereby enhancing model discrimination while reducing redundancy and noise inherent in high-dimensional financial data.

Based on the results presented, we evaluated several feature selection approaches using filters and wrappers. Among the methods evaluated, the *Information Gain Method* [14] demonstrated superior performance in terms of model stability compared to all other methods tested, showing generalization and training efficiency when applied to the intraday dataset. This method identified a compact and informative subset, selecting 7 features from the original 66.

The selected features, ranked by their Information Gain scores, include the EMA-based differences 5–3 and 7–3, their normalized counterparts, the normalized 9-period SMA, the 9–3 EMA difference, and the normalized 7-period EMA.

This ranked subset was subsequently used as the optimized input space for training the models using a GA-based hyperparameter optimization procedure. Limiting the learning space to key indicators allowed the genetic algorithm’s optimization to run more efficiently, finding parameters with better predictive capacity and less variance [15].

3.4 Implemented Classifier Architectures and Search Spaces

To benchmark metaheuristic search performance across fundamentally distinct hypothesis spaces, four classifier backbones were implemented: Random Forest (RF), Support Vector Machines (SVM), Multilayer Perceptron (MLP), and 1D Convolutional Neural Networks (1D-CNN). Figure 5 illustrates the structural decision mechanisms and layer topologies of these four implemented architectures.

The hyperparameter search space bounds $\theta \in \mathcal{S}$ for each implemented model are defined as follows:

- **RF:** Number of trees $n_{\text{estimators}} \in [50, 500]$, maximum tree depth $d_{\text{max}} \in [5, 50]$, minimum split samples $s_{\text{split}} \in [2, 20]$, minimum leaf samples $s_{\text{leaf}} \in [1, 20]$, and categorical choices for `max_features` (`{sqrt, log2, None}`) and `class_weight` (`{None, balanced}`).
- **SVM:** Soft-margin penalty $C \in [10^{-3}, 10^3]$ and kernel bandwidth $\gamma \in [10^{-5}, 10^1]$, both sampled on logarithmic scales, with linear and radial-basis-function kernels considered.

- **MLP**: Hidden-layer width $n_{\text{hidden}} \in [5, 500]$, L2 penalty coefficient $\alpha \in [10^{-8}, 10^{-2}]$, initial learning rate $\eta_0 \in [10^{-8}, 10^{-2}]$, dropout $p_{\text{drop}} \in [0, 0.10]$, and batch size in $\{16, 32, 64, 80, 112, 128\}$.
- **1D-CNN**: Filter count $n_{\text{filter}} \in [8, 128]$, kernel size $k_{\text{size}} \in [2, 5]$, dense-layer width $n_{\text{dense}} \in [16, 256]$, L2 penalty coefficient $\alpha \in [10^{-8}, 10^{-2}]$, Adam learning rate $\eta \in [10^{-8}, 10^{-2}]$, dropout $p_{\text{drop}} \in [0, 0.10]$, and batch size in $\{16, 32, 64, 80, 112, 128\}$.

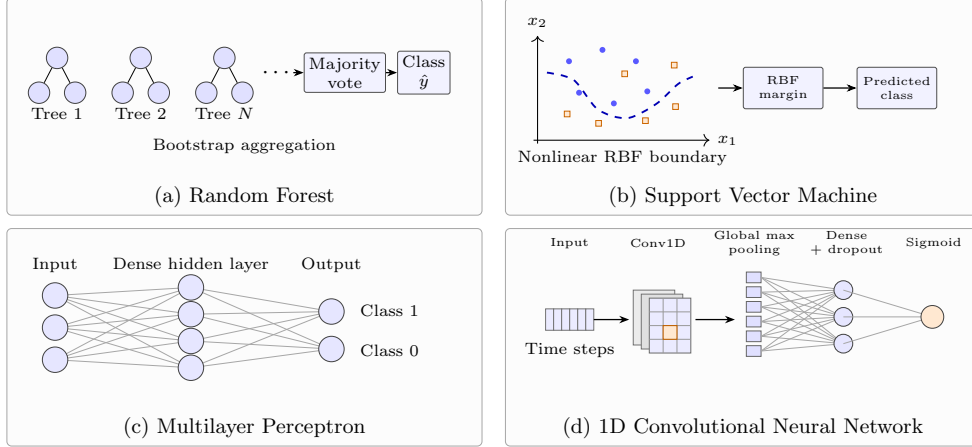


Fig. 5 Architectural topologies and structural decision mechanisms of the four candidate classifier backbones evaluated under the equal-budget metaheuristic optimization framework.

3.5 Optimizer Configuration, Search Operators, and Validation Objectives

To ensure complete experimental reproducibility, five search paradigms were evaluated: Random Search (RS), Genetic Algorithm (GA), Particle Swarm Optimization (PSO), Differential Evolution (DE), and Grey Wolf Optimizer (GWO). All optimizers operate under a strict candidate evaluation cap of $N_{\text{eval}} = 1,000$ per model and random seed. For all population-based algorithms (GA, PSO, DE, GWO), the population size is fixed to $N_{\text{pop}} = 10$ candidates evaluated across $G = 100$ generations or iterations ($10 \times 100 = 1,000$ function evaluations). RS generates $N_{\text{eval}} = 1,000$ independent uniform random samples across the search space.

Candidate vectors use the bounds stated above. Integer parameters and categorical-option indices are rounded and clipped to their admissible bounds:

$$\theta_d = \text{clip}(\text{round}(u_d), \theta_{d,\min}, \theta_{d,\max}). \quad (7)$$

For continuous parameters defined on logarithmic scales (such as C , α , and learning rates), optimizers act on $u_d \in [\log_{10}(\theta_{d,\min}), \log_{10}(\theta_{d,\max})]$ and values are decoded by

$$\theta_d = 10^{u_d}. \quad (8)$$

The specific search operators and update formulations governing each optimizer are defined as follows:

- **Genetic Algorithm (GA)**: Utilizes tournament selection ($k = 3$), uniform crossover with probability $P_c = 0.80$, and mutation with probability $P_m = 0.20$. When a gene is mutated, its perturbation scale is 0.10 of the corresponding encoded search range.
- **Particle Swarm Optimization (PSO)**: Velocity vectors $\mathbf{v}_i$ and positions $\mathbf{u}_i$ for particle i at iteration g are updated using inertia weight $w = 0.70$ and cognitive/social coefficients $c_1 = c_2 = 1.50$:

$$\mathbf{v}_i^{g+1} = w\mathbf{v}_i^g + c_1 r_1 (\mathbf{pbest}_i - \mathbf{u}_i^g) + c_2 r_2 (\mathbf{gbest} - \mathbf{u}_i^g), \quad (9)$$

$$\mathbf{u}_i^{g+1} = \text{clip}(\mathbf{u}_i^g + \mathbf{v}_i^{g+1}, 0, 1), \quad (10)$$

where $r_1, r_2 \sim U(0, 1)$.

- **Differential Evolution (DE)**: Implements the DE/best/1/bin strategy with mutation factor $F = 0.80$ and binomial crossover probability $CR = 0.90$:

$$\mathbf{v}_i^{g+1} = \mathbf{gbest}^g + F(\mathbf{u}_{r_1}^g - \mathbf{u}_{r_2}^g), \quad (11)$$

where r_1, r_2 are distinct randomly chosen population indices.

- **Grey Wolf Optimizer (GWO)**: Candidate position vectors are updated by simulating encirclement of the three leading wolves (α, β, δ):

$$\mathbf{u}_i^{g+1} = \frac{\mathbf{x}_1 + \mathbf{x}_2 + \mathbf{x}_3}{3}, \quad (12)$$

where $\mathbf{x}_1 = \alpha - \mathbf{A}_1 \cdot |\mathbf{C}_1 \cdot \alpha - \mathbf{u}_i^g|$, with analogous terms for $\mathbf{x}_2$ (β) and $\mathbf{x}_3$ (δ).

Candidate configurations are evaluated against two separated validation objectives:

- **Protocol A (MCC/ F_1 Compound Objective)**:

$$f_{\text{MCC}/F_1}(\boldsymbol{\theta}) = 0.60 \text{MCC}_{\text{val}}(\boldsymbol{\theta}) + 0.40 F_{1,\text{val}}(\boldsymbol{\theta}). \quad (13)$$

- **Protocol B (Weighted Accuracy Objective)**:

$$f_{\text{Acc}}(\boldsymbol{\theta}) = 0.40 \text{Accuracy}_{\text{train}}(\boldsymbol{\theta}) + 0.60 \text{Accuracy}_{\text{val}}(\boldsymbol{\theta}). \quad (14)$$

Upon completion of the $N_{\text{eval}} = 1,000$ search budget, the optimal candidate vector $\boldsymbol{\theta}^* = \arg \max f(\boldsymbol{\theta})$ is refit on the concatenated training and validation dataset ($N = 12,045$ bars) and evaluated exactly once on the locked test block ($N = 3,012$ bars).

The exploratory economic assessment maps an uptrend prediction to a long position and a downtrend prediction to a short position, opened at the bar open and closed at its close. Each bar is treated as one round-trip trade and is charged a fixed cost of five index points. We report total net profit, maximum drawdown, profit factor, and annualized Sharpe ratio; these quantities are therefore specific to this simplified intrabar execution rule rather than a claim of deployable live performance.

3.6 Statistical Analysis

Statistical comparisons are conducted separately within each holdout protocol and use the four classifier backbones as the groups of interest. For a given endpoint, each optimizer–seed pair defines one matched block, giving 15 blocks (five optimizers and three seeds) per protocol. We first apply the Friedman rank test to assess whether model outcomes differ across these matched blocks. When summarizing pairwise contrasts, the model with the highest marginal mean for the endpoint is compared with each of the other three models using two-sided paired Wilcoxon signed-rank tests. The resulting three p -values are adjusted together using the Holm step-down procedure, with family-wise significance level $\alpha = 0.05$.

These tests are exploratory. The blocks share one chronological test segment, and the five optimizers and three seeds do not provide 15 independent market periods. Accordingly, the tests quantify consistency of model ordering across the evaluated optimizer–seed combinations on this fixed test period; they do not establish replication across market regimes or a confirmatory difference between validation protocols. No optimizer-superiority claim is inferred from these model-level tests.

4 Results

The data partition, search budget, validation objectives, and statistical procedures are defined in Section 3. This section focuses on the empirical patterns they produce: first the deterministic reference baselines, then optimization trajectories, locked-test performance, model-level summaries, exploratory inference, and economic outcomes. Unless otherwise stated, optimizer summaries use the three common seeds; inferential blocks share the same locked chronological test interval and are interpreted accordingly.

4.1 Reference Baselines

Two deterministic reference models were evaluated on the same locked test block before interpreting the optimizer benchmark. The majority-direction baseline predicts the class that is more frequent in the training partition; its probability is fixed at that training prevalence. The default RF baseline uses the same fixed seven-feature representation and the final-refit rule (training plus validation), but no hyperparameter search. These references are not additional optimizers and do not enter the matched optimizer–seed statistical tests.

Table 2 establishes two fixed reference points before the tuned models are compared. The majority-direction rule attains accuracy 0.510 and $F_1 = 0.675$ but MCC 0, showing how the prevalent class can inflate positive-class F_1 without useful class

association. The untuned RF improves on that rule (MCC 0.210; balanced accuracy 0.604), but remains a single-backbone reference rather than a no-tuning baseline for every architecture. Thus, optimized outcomes should be interpreted relative to these limited but informative anchors.

Table 2 Locked-test reference baselines under the temporal protocol. The naive baseline selects the training-majority direction (uptrend); the default RF is refitted on training plus validation with 100 trees and no hyperparameter search.

Reference	Accuracy	Balanced accuracy	F_1	MCC	AUC-ROC
Naive majority direction	0.510	0.500	0.675	0.000	0.500
Default RF (no tuning)	0.605	0.604	0.630	0.210	0.645

The contrast in Table 2 motivates reporting MCC alongside accuracy and F_1 : no single endpoint captures both the majority-class effect and balanced predictive association. Because only RF has a default-model reference here, the table does not support claims about the benefit of tuning for the other three backbones.

4.2 Representative Search Configurations

Table 3 summarizes the search domains and one illustrative GA-selected configuration per backbone (seed 1 under the MCC/ F_1 protocol). The selected values span distinct regions of the stated domains, particularly for network width and learning rate, underscoring that a single example is not a model-family default. These four configurations are illustrative only; the benchmark conclusions below use all five optimizers and all three common seeds.

Table 3 Hyperparameter search domains and representative GA-selected configurations (seed 1, MCC/ F_1 validation objective) across the four classifier backbones.

Random Forest (RF)			Support Vector Machine (SVM)		
Hyperparameter	Search space	Selected	Hyperparameter	Search space	Selected
$n_{\text{estimators}}$	[50, 500]	58	C	$[10^{-3}, 10^3]$ (log scale)	2.089
max_depth	[5, 50]	6	γ	$[10^{-5}, 10^1]$ (log scale)	5.044
min_samples_split	[2, 20]	13	Kernel	{linear, RBF}	linear
min_samples_leaf	[1, 20]	15			
max_features	{sqrt, log2, None}	None			
class_weight	{None, balanced}	None			
Multilayer Perceptron (MLP)			1D Convolutional Neural Network (1D-CNN)		
Hyperparameter	Search space	Selected	Hyperparameter	Search space	Selected
Hidden neurons	[5, 500]	410	Filters	[8, 128]	48
Learning rate	$[10^{-8}, 10^{-2}]$ (log scale)	3.42×10^{-4}	Kernel size	[2, 5]	3
L_2 regularization	$[10^{-8}, 10^{-2}]$ (log scale)	1.66×10^{-7}	Dense neurons	[16, 256]	251
Dropout rate	[0, 0.10]	0.057	Learning rate	$[10^{-8}, 10^{-2}]$ (log scale)	3.90×10^{-6}
Batch size	{16, 32, 64, 80, 112, 128}	112	L_2 regularization	$[10^{-8}, 10^{-2}]$ (log scale)	5.70×10^{-6}
			Dropout rate	[0, 0.10]	0.026
			Batch size	{16, 32, 64, 80, 112, 128}	16

The RF example selects a shallow tree ensemble, while the neural-network examples select different widths, regularization, and learning-rate scales. These individual settings clarify what the search space permits but do not explain comparative performance by themselves; that evidence comes from the repeated convergence and locked-test summaries.

4.3 Optimization and Convergence Behaviour

The eight objective-specific plots in Figures 6–13 show mean best-so-far validation fitness across the three common seeds, without seed-level variability envelopes. Each panel has a tight y-axis based on the observed range of its five mean curves, and an inset magnifies evaluations 751–1,000. Since limits vary by model and objective, absolute fitness should be compared using the tick values rather than the relative vertical positions across panels; the plots primarily diagnose within-model search behaviour.

4.3.1 Random Forest

Figures 6 and 7 show rapid early improvement under both objectives. For MCC/ F_1 , the five trajectories approach a near-plateau early; thereafter RS continues a modest stepwise rise, while GA remains below the close DE, PSO, and GWO terminal values. Under weighted-accuracy fitness, the curves again rise sharply at the beginning and then change little, with narrow terminal separation and GWO slightly higher. Most progress therefore occurs well before the 1,000-evaluation limit. The endpoint gaps, particularly among the population methods, are descriptive and do not establish optimizer superiority.

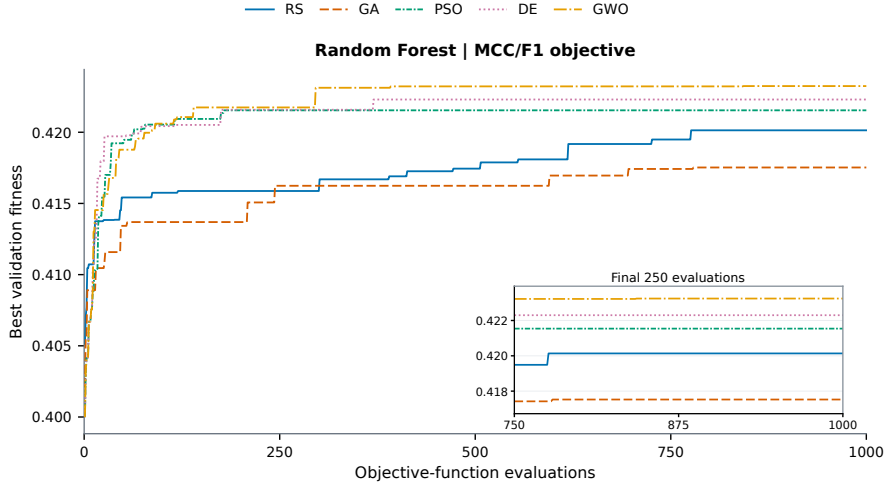


Fig. 6 Fitness evolution during equal-budget optimization of the Random Forest (RF) model under the MCC/ F_1 validation objective. Lines show mean best-so-far fitness across the three common seeds. The y-axis is tightly scaled to the observed range; the inset magnifies the final 250 objective evaluations.

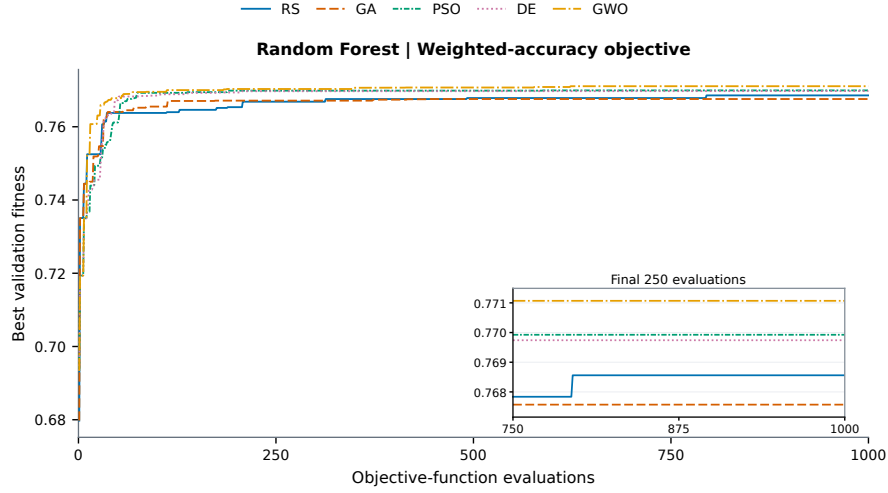


Fig. 7 Fitness evolution during equal-budget optimization of the Random Forest (RF) model under the weighted-accuracy validation objective. Lines show mean best-so-far fitness across the three common seeds. The y-axis is tightly scaled to the observed range; the inset magnifies the final 250 objective evaluations.

The RF panels differ mainly in the extent of late refinement: the MCC/F_1 trajectories retain somewhat more separation, whereas weighted-accuracy fitness ends in a tightly grouped set. Neither panel suggests a large payoff from spending the entire budget after the initial gains.

4.3.2 Support Vector Machine

Figures 8 and 9 display the strongest optimizer sensitivity. Under MCC/F_1 , GWO and PSO make larger later gains and finish above RS and DE, while GA levels off at a lower fitness. Under weighted accuracy, GWO, PSO, and RS form a close upper group, DE finishes lower, and GA is lowest. These distinctions remain visible in the terminal insets, despite a common evaluation budget.

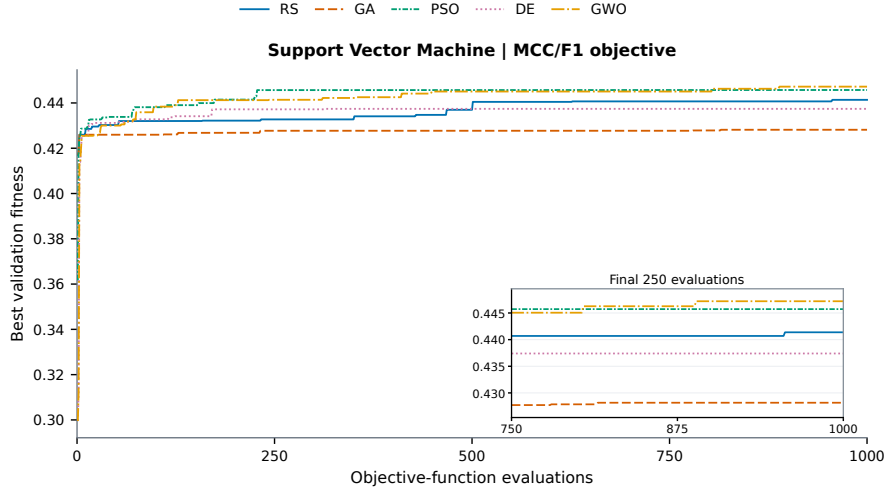


Fig. 8 Fitness evolution during equal-budget optimization of the support vector machine (SVM) under the MCC/F_1 validation objective. Lines show mean best-so-far fitness across the three common seeds. The y-axis is tightly scaled to the observed range; the inset magnifies the final 250 objective evaluations.

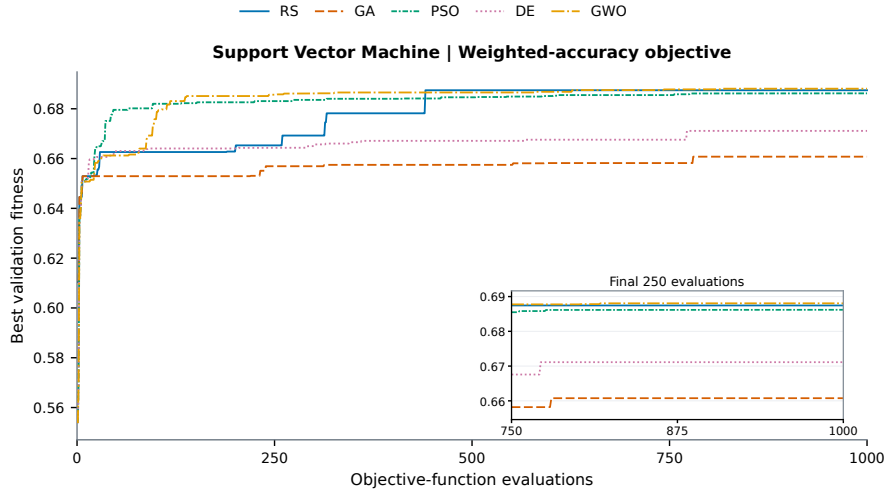


Fig. 9 Fitness evolution during equal-budget optimization of the support vector machine (SVM) under the weighted-accuracy validation objective. Lines show mean best-so-far fitness across the three common seeds. The y-axis is tightly scaled to the observed range; the inset magnifies the final 250 objective evaluations.

This search sensitivity is consistent with the wide SVM locked-test dispersion in Table 5. However, validation ranking does not determine the test ranking: under MCC/F_1 , GA has the largest mean test MCC (0.290), although its validation-fitness trajectory finishes below GWO and PSO. Under weighted accuracy, GWO leads both the validation trajectory and the test-accuracy means, an alignment observed here rather than a general guarantee.

4.3.3 Multilayer Perceptron

Figures 10 and 11 reveal objective-dependent behaviour. With MCC/F_1 fitness, DE, PSO, GA, and GWO largely flatten during the later budget, whereas RS continues to improve and finishes above the other curves. That visual validation advantage does not carry through to locked-test MCC : RS averages 0.106, compared with 0.305 for GWO and 0.302 for DE in Table 4. Under weighted-accuracy fitness, all five trajectories become tightly clustered after their early gains and remain nearly indistinguishable near termination.

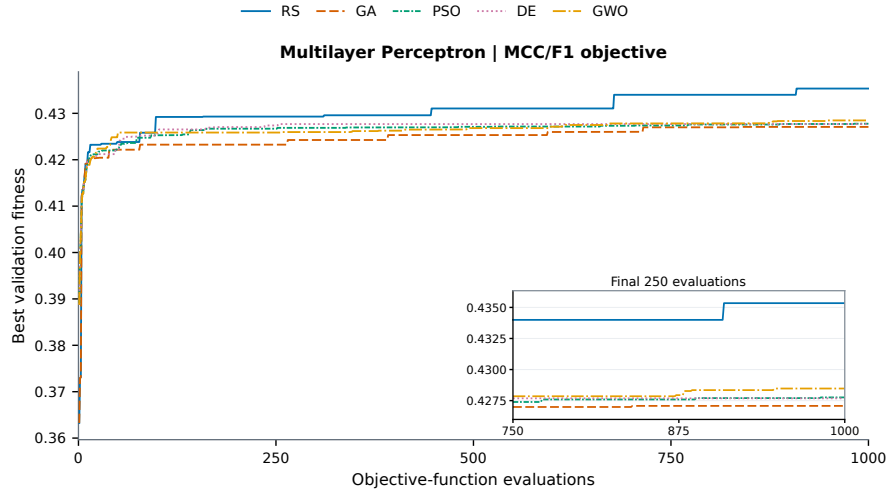


Fig. 10 Fitness evolution during equal-budget optimization of the multilayer perceptron (MLP) under the MCC/F_1 validation objective. Lines show mean best-so-far fitness across the three common seeds. The y-axis is tightly scaled to the observed range; the inset magnifies the final 250 objective evaluations.

The MLP curves show why visually similar terminal validation fitness must not be treated as identical generalization: under weighted accuracy, the locked-test means still span optimizer-specific values, while the MCC/F_1 panel’s late RS advantage is paired with the lowest RS test MCC . These descriptive contrasts indicate that the objective shapes the selected search region, but do not establish why a particular configuration generalizes better.

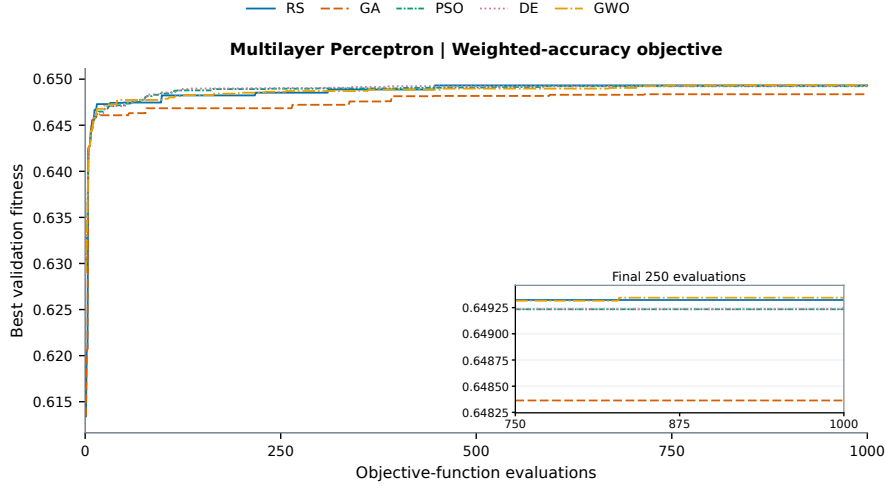


Fig. 11 Fitness evolution during equal-budget optimization of the multilayer perceptron (MLP) under the weighted-accuracy validation objective. Lines show mean best-so-far fitness across the three common seeds. The y-axis is tightly scaled to the observed range; the inset magnifies the final 250 objective evaluations.

4.3.4 One-Dimensional Convolutional Neural Network

Figures 12 and 13 show rapid early progress followed by a stable tail. Under MCC/F_1 , PSO and GWO end at the upper edge of a compact group; GA and DE level off slightly lower, and RS lies between them. Under weighted accuracy, trajectories are again close after the initial rise, with GWO finishing marginally above the others. Relative to SVM, optimizer separation is narrower, especially for the weighted-accuracy trajectory.

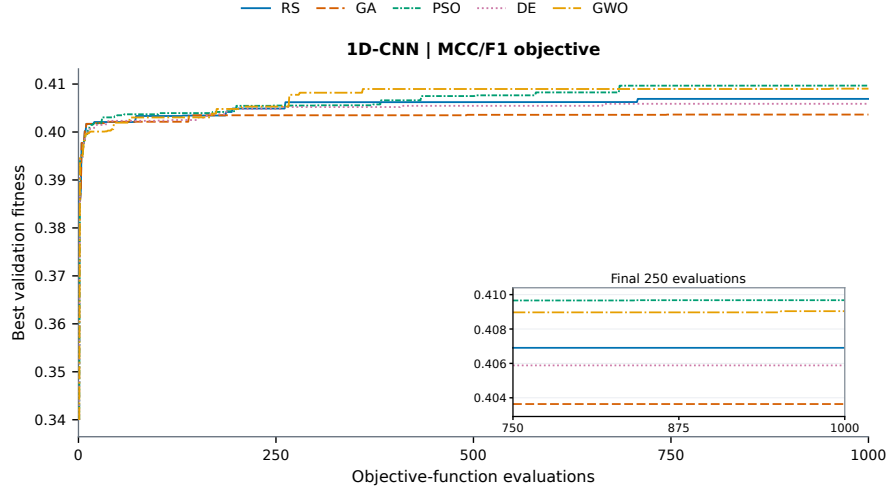


Fig. 12 Fitness evolution during equal-budget optimization of the one-dimensional convolutional neural network (1D-CNN) under the MCC/F_1 validation objective. Lines show mean best-so-far fitness across the three common seeds. The y-axis is tightly scaled to the observed range; the inset magnifies the final 250 objective evaluations.

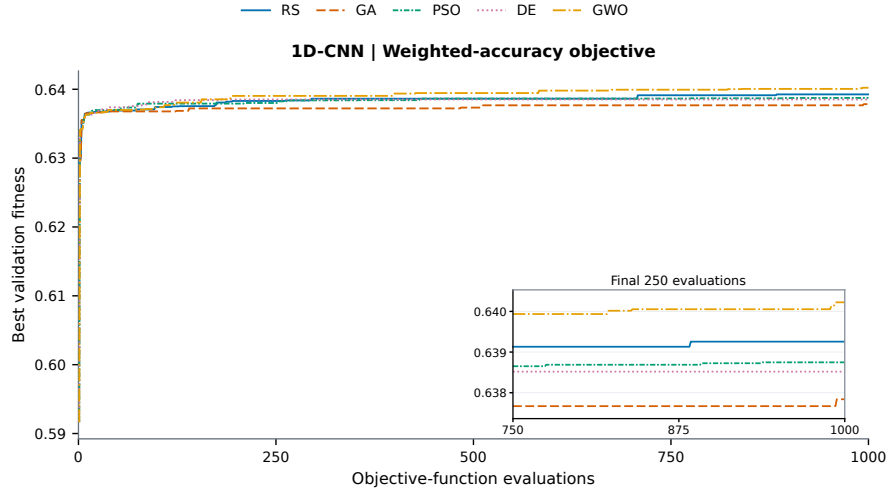


Fig. 13 Fitness evolution during equal-budget optimization of the one-dimensional convolutional neural network (1D-CNN) under the weighted-accuracy validation objective. Lines show mean best-so-far fitness across the three common seeds. The y-axis is tightly scaled to the observed range; the inset magnifies the final 250 objective evaluations.

The test table qualifies these small validation gaps. Under MCC/F_1 , PSO has the largest mean test MCC (0.269), while GWO’s test MCC is 0.139 despite its near-leading validation trajectory. Under weighted accuracy, GWO also has the largest mean test accuracy (0.637), but the other four means are close (0.634–0.636). These are descriptive contrasts across three seeds, not evidence that a convergence curve identifies a statistically superior optimizer. Validation-fitness convergence measures search behaviour on development data; the locked chronological test block provides the relevant generalization comparison below.

4.4 Optimizer-Level Performance Under Equal Budget

Convergence describes selection on validation fitness, not performance on the held-out period. Table 4 therefore compares the locked-test endpoint aligned with each validation protocol: MCC for MCC/F_1 selection and accuracy for weighted-accuracy selection. Entries are means over three common seeds under the same 1,000-evaluation budget; they are descriptive summaries, not tests of optimizer superiority.

Table 4 Optimizer-level locked-test primary metrics under equal budget. Each cell is a mean over three common seeds; boldface denotes the largest mean within a model and protocol. RS: Random Search; GA: Genetic Algorithm; PSO: Particle Swarm Optimization; DE: Differential Evolution; GWO: Grey Wolf Optimizer.

Model	RS	GA	PSO	DE	GWO
Panel A: MCC/F_1 fitness; locked-test MCC					
RF	0.281	0.293	0.293	0.292	0.288
SVM	0.105	0.290	0.255	0.187	0.232
MLP	0.106	0.290	0.299	0.302	0.305
1D-CNN	0.179	0.268	0.269	0.267	0.139
Panel B: weighted-accuracy fitness; locked-test accuracy					
RF	0.604	0.611	0.604	0.600	0.602
SVM	0.588	0.557	0.571	0.584	0.593
MLP	0.649	0.649	0.652	0.652	0.652
1D-CNN	0.636	0.634	0.635	0.634	0.637

The table shows an optimizer-by-model interaction rather than a common winner. Under MCC/F_1 , RF is comparatively insensitive among GA, PSO, and DE (0.293, 0.293, and 0.292), whereas its RS mean is 0.281. SVM is more sensitive: GA reaches 0.290, but RS reaches only 0.105; for MLP, GWO is highest at 0.305 while RS is again 0.106. The 1D-CNN is tightly grouped across GA, PSO, and DE (0.267–0.269), but lower for RS and particularly GWO. Under weighted-accuracy selection, the five MLP means are tightly clustered (0.649–0.652); the SVM range is wider (0.557–0.593), while RF spans 0.600–0.611 and 1D-CNN spans 0.634–0.637. Thus, optimizer sensitivity depends on both the backbone and the selected endpoint. With only three common seeds, the within-row maxima are not evidence of a universally best optimizer; the aggregated model comparison asks instead whether these patterns alter the backbone-level picture.

4.5 Model-Level Aggregation Across Optimizers

Table 5 and Figure 14 jointly summarize the locked-test outcomes over 15 optimizer-seed blocks per backbone and protocol. The table retains dispersion across those blocks; the figure makes the change in model ordering between validation objectives easier to see. These are aggregate descriptions of the evaluated search procedures, not 15 independent market replications.

Table 5 Locked-test predictive outcomes by backbone and validation protocol. Entries are mean $\pm$ standard deviation over 15 optimizer-seed blocks per row (five optimizers $\times$ three common seeds).

Panel A: MCC/F_1 validation fitness			
Model	Accuracy	MCC	F_1
RF	0.645 ± 0.004	0.289 ± 0.008	0.664 ± 0.006
SVM	0.604 ± 0.060	0.214 ± 0.132	0.602 ± 0.122
MLP	0.630 ± 0.070	0.261 ± 0.141	0.639 ± 0.071
1D-CNN	0.613 ± 0.056	0.224 ± 0.120	0.597 ± 0.166
Panel B: Weighted-accuracy validation fitness			
Model	Accuracy	MCC	F_1
RF	0.604 ± 0.005	0.208 ± 0.009	0.624 ± 0.004
SVM	0.578 ± 0.028	0.159 ± 0.053	0.600 ± 0.071
MLP	0.651 ± 0.003	0.301 ± 0.005	0.658 ± 0.004
1D-CNN	0.635 ± 0.002	0.271 ± 0.005	0.639 ± 0.007

In Panel A of Table 5, RF has the highest aggregate mean for accuracy, MCC, and F_1 , and its standard deviations are small (0.004, 0.008, and 0.006). SVM, MLP, and 1D-CNN show appreciably larger dispersion, especially for MCC/ F_1 selection. Panel B reverses the leading backbone: MLP has the highest mean across all three metrics and low dispersion; 1D-CNN is also competitive, while RF’s mean accuracy and MCC are lower than in Panel A. The pattern suggests that the validation objective changes which model/configuration regions the search favours; it does not establish that either objective is globally superior.

Figure 14 makes this objective-dependent ordering visible: RF leads the MCC/ F_1 panel-level means, whereas MLP leads under weighted-accuracy selection, with 1D-CNN close behind MLP on weighted-accuracy test accuracy. The figure’s grouped bars show means, while Table 5 supplies the dispersion needed to judge stability. Accuracy and MCC are complementary here: a change in overall correct classifications need not preserve the model’s class-balanced association, so neither panel should be reduced to one endpoint. These differences are meaningful as descriptions of this locked interval, but the single chronological test period limits claims about future market regimes.

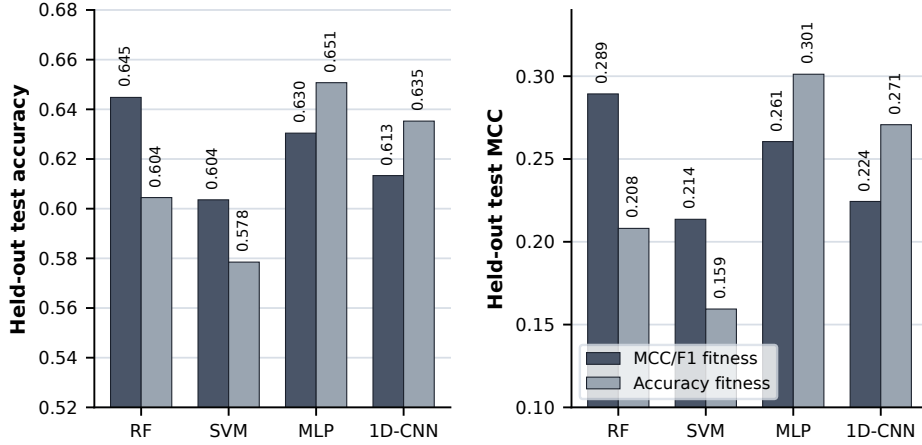


Fig. 14 Mean locked-test accuracy and MCC for four model families under the two validation objectives, aggregated over 15 optimizer–seed blocks per model and protocol.

4.6 Exploratory Statistical Evidence

Table 6 reports exploratory Friedman comparisons of the four backbones within each protocol. In each of the 15 blocks (five optimizers crossed with three seeds), the backbones are ranked on the protocol-aligned locked-test endpoint; the Friedman statistic assesses whether those within-block rankings are uniform. The stacked panels keep each global test beside the corresponding backbone mean ranks.

Under MCC/F_1 , the Friedman test detects differences in matched-block ordering ($\chi^2 = 22.84$, $p = 4.36 \times 10^{-5}$). RF has the largest marginal mean MCC in Table 5, while MLP has the lowest mean rank in Table 6; this is not contradictory. A marginal mean aggregates score magnitudes across blocks, whereas the mean rank first orders the backbones separately within each block and then averages those ranks. A model can therefore rank consistently well without having the largest pooled score magnitude. Under weighted accuracy, MLP leads both the mean accuracy and mean rank; the reported Holm-adjusted contrasts against RF, SVM, and 1D-CNN are significant ($p_{\text{Holm}} \approx 0.00194$) within this analysis.

These results remain exploratory. The 15 blocks are optimizer–seed combinations on one shared chronological market interval, not 15 independent market replications. The tests describe consistency across the evaluated search combinations; they do not establish replication across regimes or a confirmatory difference between validation objectives. Their interpretation should remain paired with the descriptive dispersion in Table 5.

4.7 Exploratory Economic Evaluation

Table 7 and Figure 15 compare return and downside risk under the fixed intrabar execution rule. The table reports net profit and maximum drawdown with dispersion, alongside mean profit factor and annualized Sharpe; the figure places mean return

Table 6 Exploratory Friedman tests for protocol-aligned locked-test endpoints across four backbones. Each test uses 15 matched optimizer–seed blocks from the same chronological test interval. Mean ranks are block-wise; lower is better.

Panel A: MCC/ F_1 fitness				
Endpoint	n	χ^2	p	
Test MCC	15	22.84	4.36×10^{-5}	
Mean ranks (lower is better)				
Model	RF	SVM	MLP	1D-CNN
Mean rank	2.07	2.73	1.53	3.67

Panel B: Weighted-accuracy fitness				
Endpoint	n	χ^2	p	
Test accuracy	15	42.92	2.56×10^{-9}	
Mean ranks (lower is better)				
Model	RF	SVM	MLP	1D-CNN
Mean rank	3.13	3.87	1.00	2.00

against mean drawdown. This joint view is important because total profit alone does not represent a complete return–risk profile.

Table 7 Exploratory economic outcomes under the fixed intrabar execution rule, aggregated over 15 optimizer–seed blocks per model and protocol. Net profit and maximum drawdown (Max. DD) are points and reported as mean $\pm$ standard deviation; PF and annualized Sharpe are means.

Protocol	Model	Net profit	Max. DD	PF	Sharpe
MCC/ F_1 fitness	RF	26,465 $\pm$ 1,732	−2,126 $\pm$ 141	1.48	12.50
	SVM	17,017 $\pm$ 29,437	−9,823 $\pm$ 17,123	1.39	8.52
	MLP	22,110 $\pm$ 20,556	−5,503 $\pm$ 12,984	1.43	10.62
	1D-CNN	19,969 $\pm$ 17,574	−5,991 $\pm$ 10,469	1.37	7.67
Weighted accuracy fitness	RF	21,973 $\pm$ 1,336	−1,866 $\pm$ 365	1.38	11.95
	SVM	11,440 $\pm$ 13,350	−5,241 $\pm$ 1,961	1.21	4.18
	MLP	27,737 $\pm$ 1,069	−2,141 $\pm$ 112	1.51	13.07
	1D-CNN	26,861 $\pm$ 1,924	−2,494 $\pm$ 62	1.49	9.71

The return–risk comparison in Figure 15 shows the protocol-dependent change in mean profit and drawdown, while Table 7 adds the other endpoints. Under MCC/ F_1 , RF combines the largest mean net profit (26,465 points) with PF 1.48, Sharpe 12.50, and a comparatively small mean drawdown (−2,126 points) with SD 141. Under weighted accuracy, MLP has the largest mean profit (27,737 points), PF 1.51, and

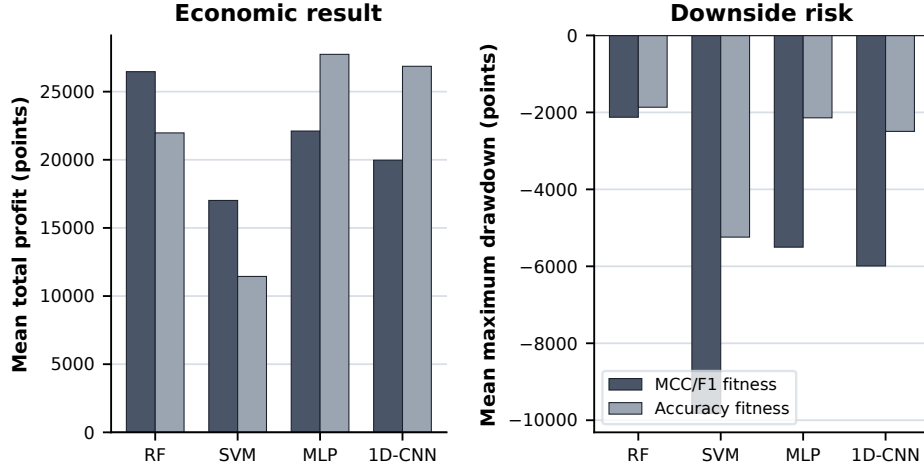


Fig. 15 Mean total profit and mean maximum drawdown across 15 optimizer–seed blocks per model and validation protocol. Both axes are in points; a drawdown closer to zero indicates a smaller loss peak-to-trough.

Sharpe 13.07; RF instead has the least negative mean drawdown (-1,866 points). The metric leader therefore depends on which component of return and risk is prioritized, and the most profitable mean is not by itself a complete ranking. These backtest summaries are exploratory under the specified rule and costs, not a trading recommendation or a transaction-cost-complete assessment.

Appendix A provides complementary views. Figures 16 and 17 show how the optimizer-by-model pattern depends on protocol; Figure 18 exposes seed-level MCC dispersion hidden by cell means. Figure 19 summarizes optimizer ordering descriptively, Figure 20 displays the performance–candidate-fit-time trade-off, and Figure 21 gives a normalized multi-metric overview. These supplementary plots help locate patterns but do not replace the tables or extend the inferential scope of the single-period experiment.

5 Discussion

The purpose of this benchmark is not to replace the practical pipeline of the blinded precursor, but to determine which conclusions remain when the temporal split, optimizer budget, validation objective, and economic endpoint are made explicit. The resulting evidence separates a model’s average locked-test outcome, its sensitivity to the optimizer–seed block, and its simplified trading consequence. This distinction matters for intraday financial prediction, where a change in selection criterion can alter a model ranking without demonstrating a stable change in market-wide forecasting ability.

5.1 What the Controlled Comparison Establishes

The present design preserves the precursor’s technical-indicator workflow and fixed seven-feature representation, but changes the comparison unit. Rather than reporting a GA-selected configuration after randomized cross-validation, every backbone is evaluated with RS, GA, PSO, DE, and GWO under the same 1,000-candidate allowance and the same chronological train/validation/test partition. Random Search is therefore a budget-matched reference against which directed population methods can be interpreted [34]. The common test segment also prevents final performance values from selecting hyperparameters.

This control improves internal comparability, but it narrows the claim that the data can support. The evidence concerns four classifier families on one five-minute WIN period, conditional on an inherited input representation. It does not show that the seven indicators would be selected again in another training period, nor that an optimizer with a favourable result here will retain that advantage under another market regime. The contribution is consequently a controlled experimental frame, not a universal ordering of metaheuristics.

5.2 Predictive Performance Depends on the Validation Objective

The locked-test summaries in Table 5 show a clear reversal at the model level. Under the compound MCC/F_1 objective, RF has the largest marginal means for accuracy, MCC, and F_1 , and its small standard deviations indicate comparatively low sensitivity to the evaluated optimizer–seed blocks. Under weighted validation accuracy, MLP is first on all three reported predictive measures, while 1D-CNN is also stable but remains below MLP in mean accuracy and MCC. The change in ordering is substantive: selecting for validation accuracy does not simply rescale the MCC/F_1 result.

MCC is useful because it summarizes the association between predicted and observed binary classes beyond raw accuracy [24]; however, optimizing an imbalance-aware criterion cannot guarantee the best outcome for every backbone and search trajectory. Conversely, the MLP advantage under weighted accuracy is concentrated in a low-variance regime, whereas MLP, SVM, and 1D-CNN outcomes under MCC/F_1 fitness vary much more across the evaluated blocks. A reported mean must therefore be accompanied by its dispersion and by the selection rule that generated it.

The rank analysis reinforces this caution. In the MCC/F_1 protocol, RF has the highest marginal mean MCC, but MLP has the lowest mean Friedman rank (Table 6). This is not a contradiction: means weight score magnitude, whereas ranks summarize ordering within each matched optimizer–seed block. Under weighted accuracy, both summaries place MLP first, and its Holm-adjusted contrasts against all other backbones are significant. Because all 15 blocks share one locked chronological segment, these results demonstrate separation within the evaluated design rather than independent replication across time.

5.3 Optimizer Trajectories Are Diagnostic, Not a Universal League Table

Figures 6–13 index progress by candidate evaluations rather than by optimizer-specific generation counts. Each line shows the mean trajectory over the three common seeds; variability envelopes are omitted to keep closely spaced optimizer curves legible. In each panel, the main y-axis is tightly bounded by the observed range of the five mean trajectories, with a small margin, and an inset magnifies the final 250 evaluations. Because limits are panel-specific, visual heights should not be compared across panels without reading their y-axis scales. Early gains indicate how quickly an optimizer locates useful regions of a search space, whereas later plateaus show the incremental benefit of additional evaluations under the fixed budget. Convergence in validation fitness does not establish equivalent generalization, so the locked-test values and their dispersion in Table 5 remain necessary for interpretation.

Search behaviour interacts with the backbone and the objective. A method that improves early for RF need not be the most reliable procedure for MLP or 1D-CNN, whose mixed discrete–continuous hyperparameter spaces have different response surfaces. The article therefore does not claim a universal winner among RS, GA, PSO, DE, and GWO. A stronger optimizer-level claim would require a dedicated analysis across more seeds and temporal replications, with the optimizer itself as the inferential factor.

5.4 Predictive Quality and Economic Outcome Are Distinct Endpoints

Table 7 makes the distinction between predictive evaluation and the exploratory trading rule concrete. Under MCC/ F_1 fitness, RF combines the largest mean net profit with the smallest average drawdown and the lowest drawdown dispersion. Under weighted accuracy, MLP has the largest mean net profit, profit factor, and annualized Sharpe ratio, while RF retains the least negative average drawdown. Thus, the preferred choice depends on whether the decision criterion prioritizes return, drawdown, or consistency; it cannot be inferred from accuracy alone.

These values are deliberately reported as points under a fixed intrabar long/short rule with a five-point round-trip cost. They are not an investable-strategy claim: the evaluation does not model a full transaction-cost schedule, market impact, delayed execution, position sizing, or a live signal-selection policy. Its role is diagnostic: it shows that converting a directional label to trades can change the practical ranking of model families, and that economic reporting should accompany predictive metrics.

5.5 Limitations and Research Agenda

Four limitations delimit the present evidence. First, only three seeds are common to every model–optimizer cell; standard deviations and rank tests describe the evaluated blocks, not repeated market samples. Second, the locked test covers one instrument and one chronological segment, so regime sensitivity and cross-market transfer remain unknown. Third, the Information-Gain mask was inherited from the blinded precursor

rather than re-estimated on the current training segment; all conclusions are conditional on that representation. Fourth, equalizing the budget at 1,000 evaluations establishes fairness at one computational scale but does not characterize larger-budget behaviour or wall-clock efficiency.

The next study should preserve the current leakage-aware split and add a pre-registered common 20-seed design across multiple non-overlapping market periods. It should extend the present naive and default-RF references to no-tuning baselines for every backbone; archive every candidate evaluation; report compute time; and test optimizer differences separately from model differences. Re-estimating feature selection within each training window would further distinguish the effect of representation from hyperparameter search. These additions would turn the current transparent benchmark into confirmatory evidence without weakening its temporal protocol.

6 Conclusion

This study transforms a GA-based conference precursor into a controlled Neural Computing and Applications benchmark for neural intraday trend classification. It preserves the predecessor’s practical insight—a compact, Information-Gain-selected technical representation can support metaheuristic hyperparameter search—but replaces randomized validation and a single-optimizer design with a chronological dual-fitness protocol, five budget-matched optimizers, and locked predictive and economic assessment.

The current evidence indicates that the choice between compound MCC/F_1 fitness and accuracy fitness produces configuration-dependent changes in both held-out accuracy and MCC. Within the accuracy-fitness protocol, MLP is the strongest model in the exploratory predictive analysis and leads mean total profit, although RF has the smallest mean drawdown. These results reinforce the central methodological lesson: selection objective, reported metric, and economic utility must be treated as related but distinct quantities.

The manuscript deliberately avoids a confirmatory claim about a universal best fitness function because only three paired seeds are currently available across every cell. Completing the common 20-seed design is the necessary next step. The resulting framework nevertheless provides a transparent and reproducible basis for future optimizer comparisons in non-stationary financial time series.

A Supplementary benchmark visualizations

This appendix provides supplementary descriptive views of the locked-test results and computational effort. Unless otherwise stated, cell summaries and plotted points use the three common seeds. The performance–cost figure uses summed recorded fit times for cache-miss candidates; it excludes cache hits and is not end-to-end wall-clock search time.

Figures 16 and 17 provide cell-level views of the protocol-aligned locked-test metric across backbone and optimizer. Their purpose is to expose interactions that are compressed in the optimizer-level table, not to add a new statistical test.

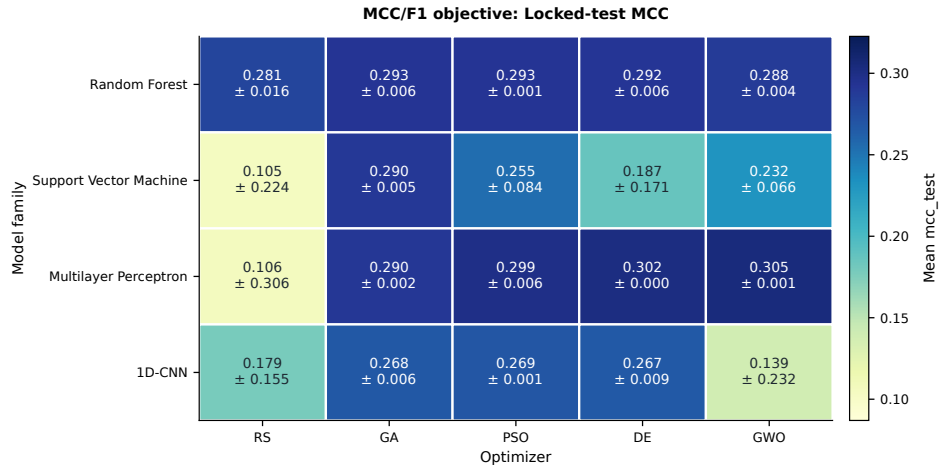


Fig. 16 Locked-test MCC by classifier backbone and optimizer after selection under the MCC/ F_1 validation objective. Each cell reports the mean and standard deviation across the three common seeds; color encodes the mean.

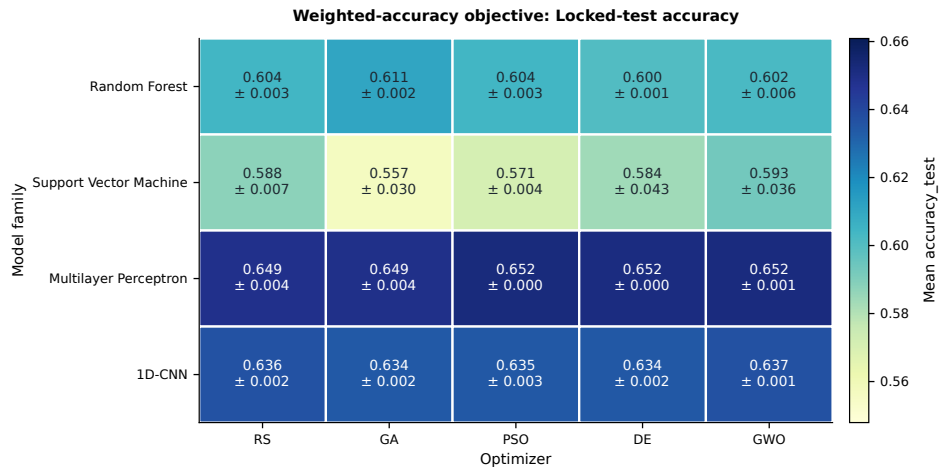


Fig. 17 Locked-test accuracy by classifier backbone and optimizer after selection under the weighted-accuracy validation objective. Each cell reports the mean and standard deviation across the three common seeds; color encodes the mean.

Across the heatmaps, RF is comparatively uniform across optimizers under MCC/ F_1 , while SVM, MLP, and 1D-CNN show more optimizer-dependent MCC patterns. Under weighted-accuracy selection, MLP forms the strongest and most consistent row; 1D-CNN is competitive but remains below it, while SVM shows wider optimizer variation. These cell patterns agree with Table 4: no optimizer preserves the same relative position for every backbone and protocol. Because each cell averages only three seeds, the colors are descriptive and should be read alongside the displayed dispersion.

Figure 18 shows how much seed-level MCC variation is hidden by the cell means, while Figure 19 summarizes optimizer ordering across the 12 matched backbone–seed blocks. The two views address complementary questions: dispersion within model–optimizer cells and variation in optimizer rank across backbones.

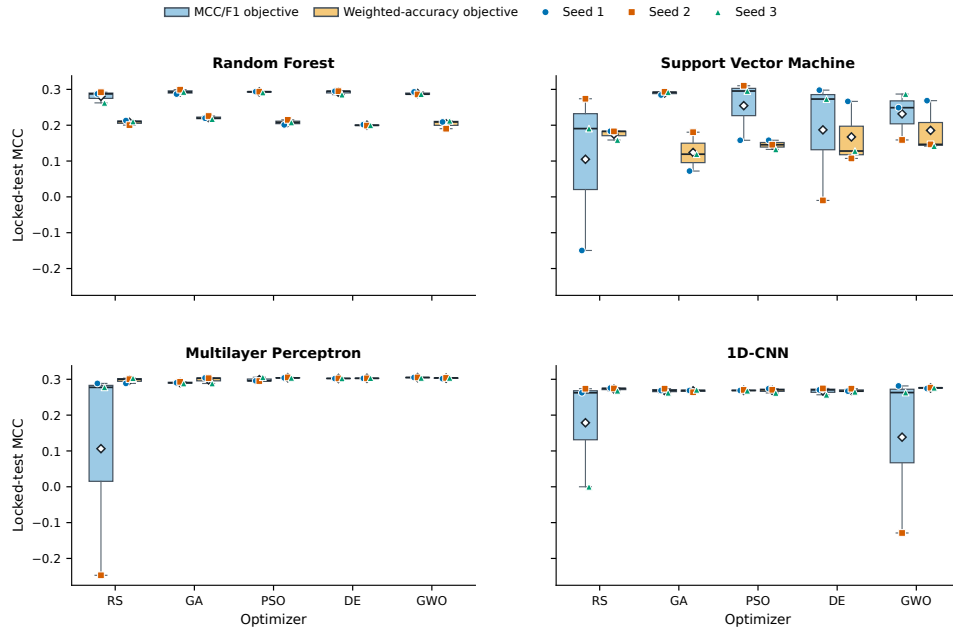


Fig. 18 Locked-test MCC across optimizer–seed outcomes for each backbone under both validation objectives. Boxes summarize the three common seeds, individual markers show seed-level values, and diamonds mark the mean. The small number of seeds makes these descriptive displays rather than inferential evidence.

The boxplots show tight RF seed clusters but visibly broader MCC variation for SVM and for selected MLP and 1D-CNN optimizer cells, including Random Search. This supports the dispersion reported in Table 5 and cautions against reading a three-seed mean as a stable cell-level estimate. The optimizer-rank plot changes its central ordering between protocols and has substantial block-level spread; it is therefore a descriptive check on heterogeneity, not an optimizer significance test or a universal league table.

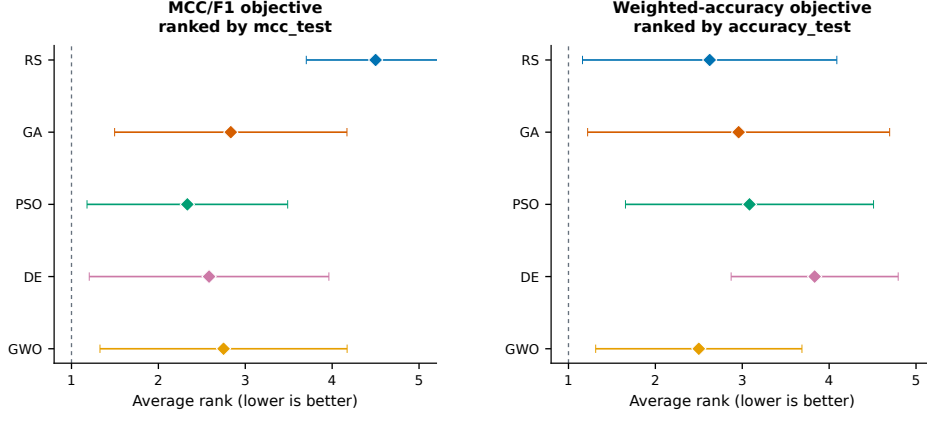


Fig. 19 Descriptive average ranks of the five optimizers across 12 matched backbone–seed blocks (four backbones and three common seeds). Rankings use locked-test MCC under the MCC/ F_1 protocol and locked-test accuracy under the weighted-accuracy protocol. Error bars denote one standard deviation of block-level ranks; lower average rank is better. No significance test is implied by this visualization.

Figure 20 places predictive score against recorded candidate-fit effort, and Figure 21 gives a normalized overview of predictive and computational dimensions. The former helps identify score–effort trade-offs across backbone–optimizer combinations; the latter is included only as a compact visual summary because normalization obscures the original units.

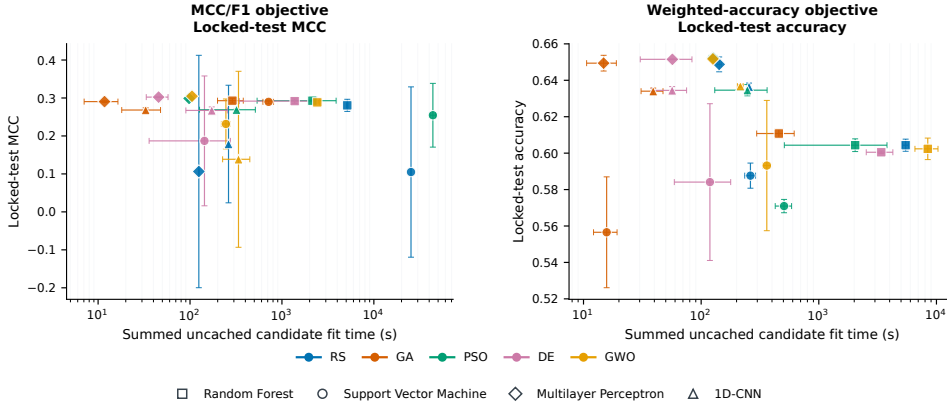


Fig. 20 Locked-test performance versus summed model-fit time for cache-miss candidate evaluations. The left panel uses test MCC for the MCC/ F_1 protocol and the right panel uses test accuracy for the weighted-accuracy protocol. Point color identifies the optimizer and marker shape identifies the backbone; points show means and error bars show one standard deviation across the three common seeds. The horizontal axis is cumulative recorded fit time, not end-to-end wall-clock search time.

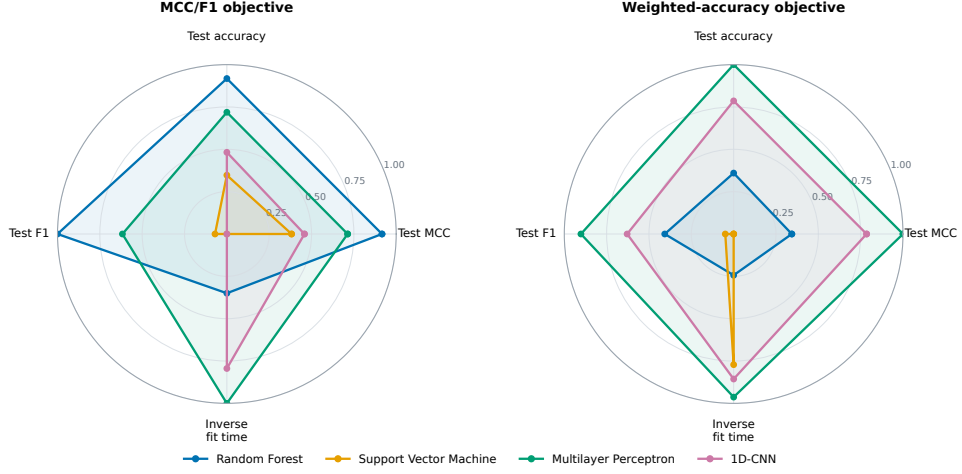


Fig. 21 Complementary radar summary of test MCC, test accuracy, test F_1 , and inverse candidate-fit time by backbone under each validation objective. Values are min-max normalized across the eight backbone-protocol averages; higher values indicate better results on each axis. Candidate-fit time is log-transformed before inversion and normalization.

The combinations in Figure 20 occupy different performance-effort positions, so the apparent advantage depends on whether predictive score or candidate-fit effort is prioritized. Its horizontal coordinate is summed uncached fit time and must not be interpreted as total optimizer wall-clock cost. In Figure 21, the profile shapes also change across protocols and metrics, reinforcing that no one scalar summary represents all outcomes. The normalized radar is consequently complementary only; quantitative comparisons should use the original-scale results in the main tables.

Declarations

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Conflict of Interest

The authors declare that they have no conflict of interest or competing financial interests.

Ethical Approval

This article does not contain any studies with human participants or animals performed by any of the authors.

Data and Code Availability

The datasets generated and analyzed during the current study, along with the complete source code for all optimization algorithms, benchmark scripts, and evaluation pipelines, are available in the project repository upon reasonable request.

Author Contributions

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